ALERT meeting

Kalman Filter updates

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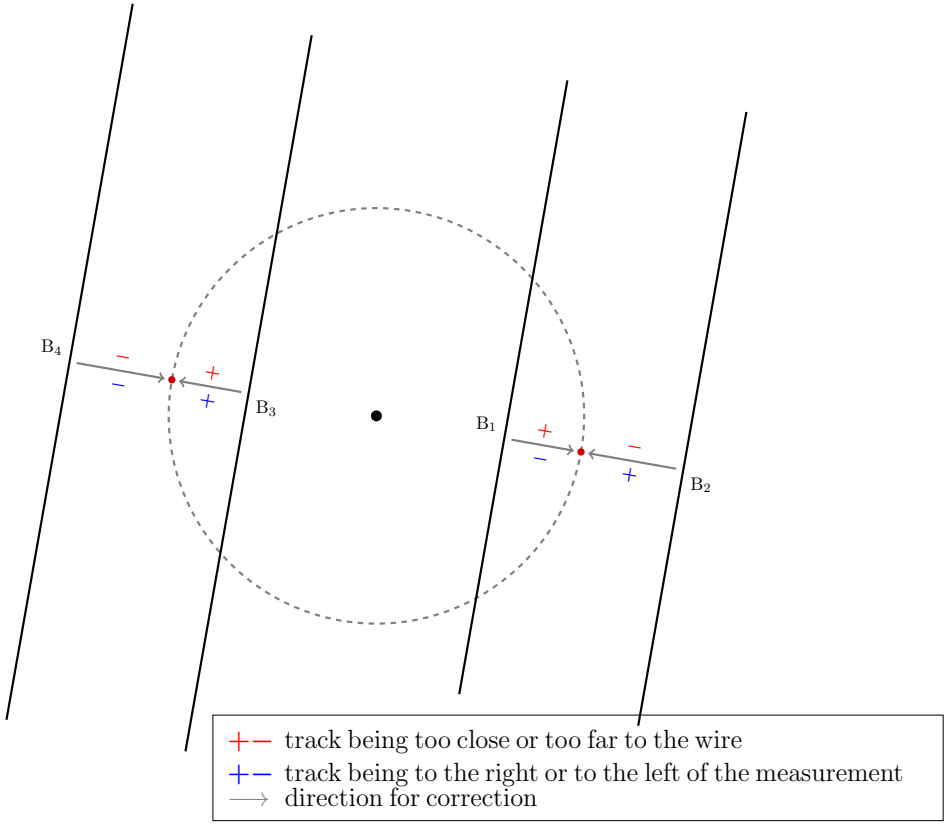
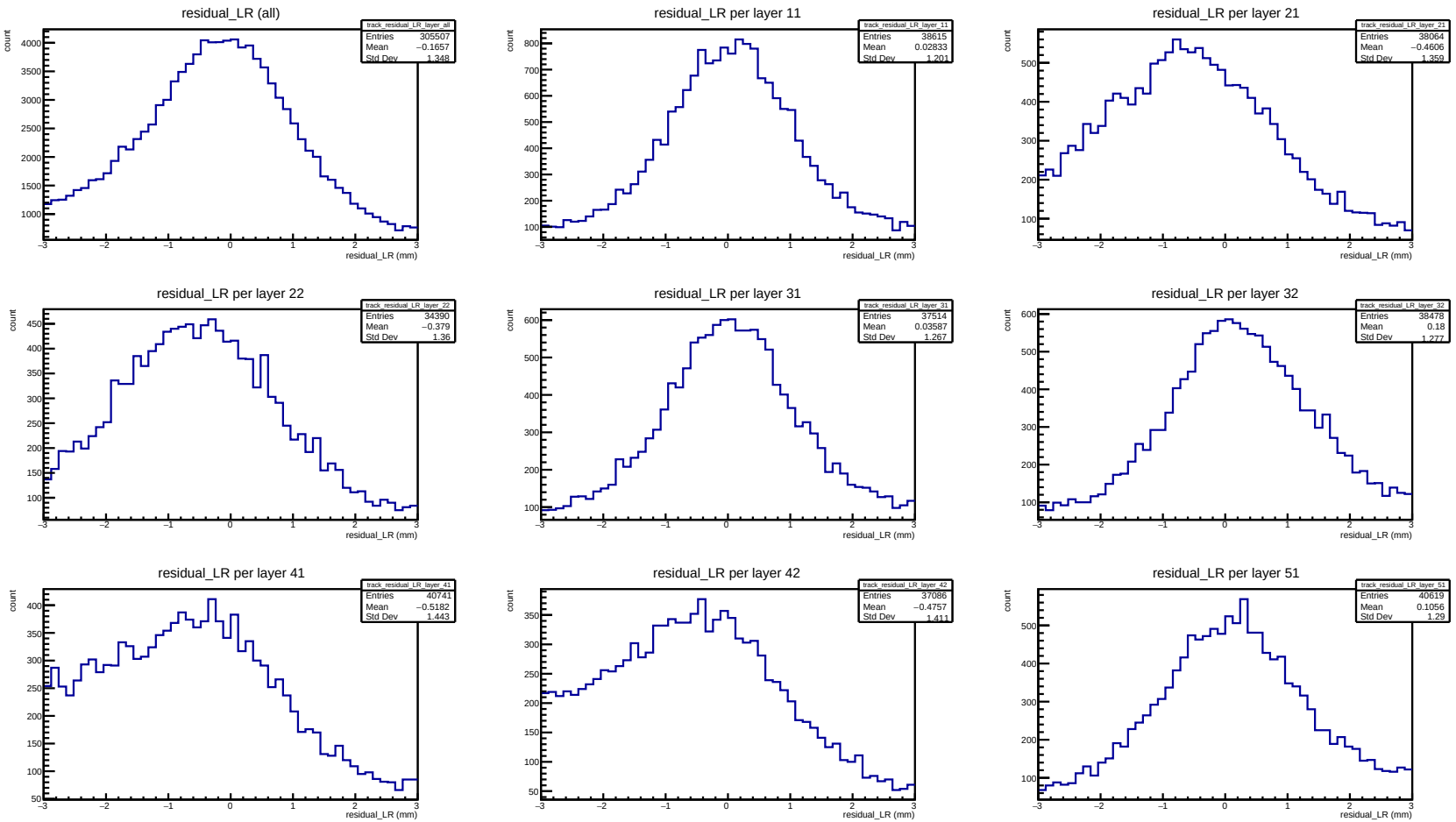
Updates - AHDC alignement

- Procedure:
  1. We looked at elastic events : electron + track (here: collection of hits only)
  2. We computed the theoretical momentum vector of the track $\vec{p}$
  3. We made the propagation without doing any fit (i.e no KF correction)
  4. We looked at residual_LR (left-right) $\longrightarrow$ sensitive to the alignment



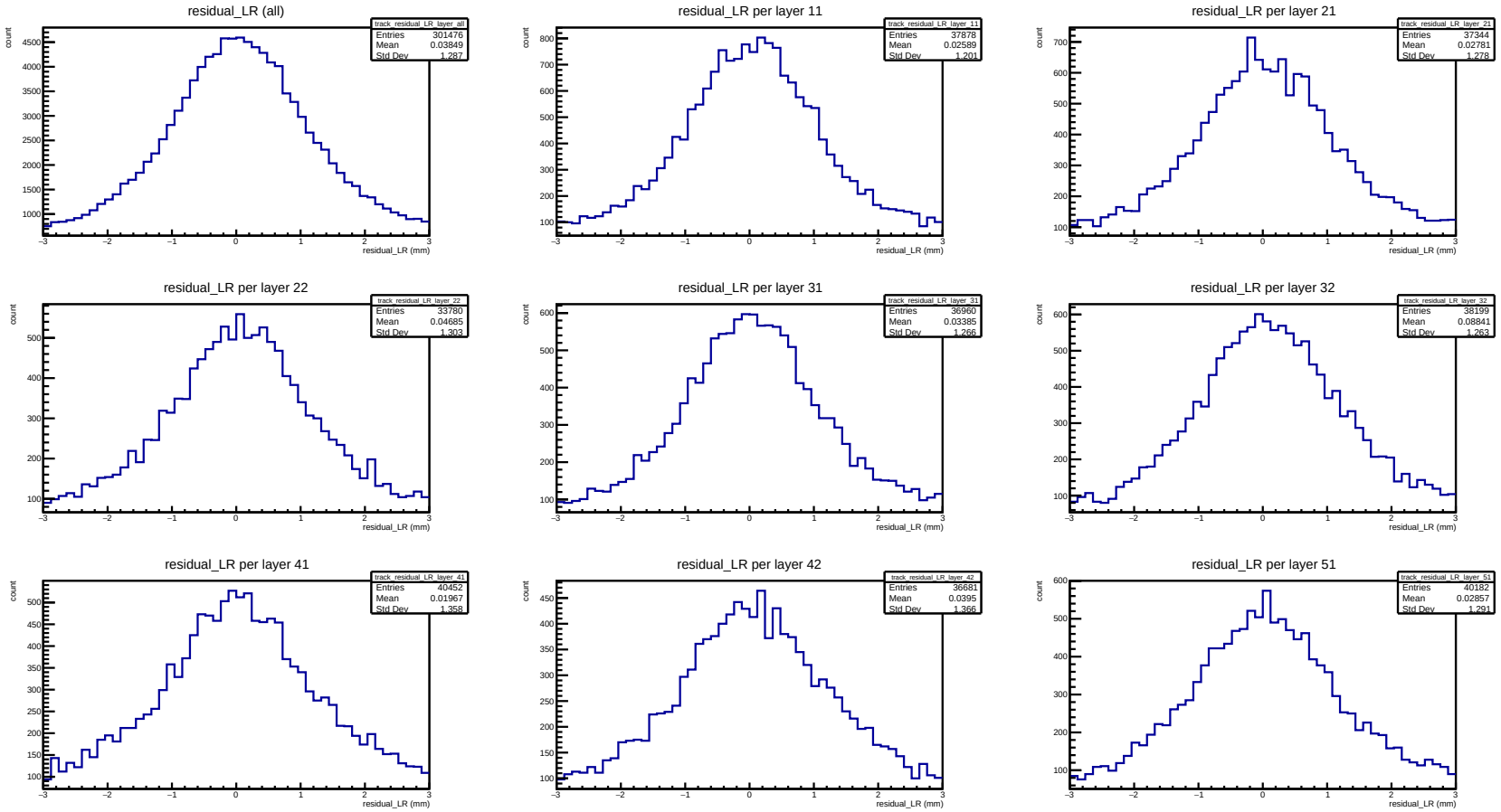
Updates - AHDC alignement

- Residual LR with no alignment (no fit)
- Even layer number need to be rotated to the phi +



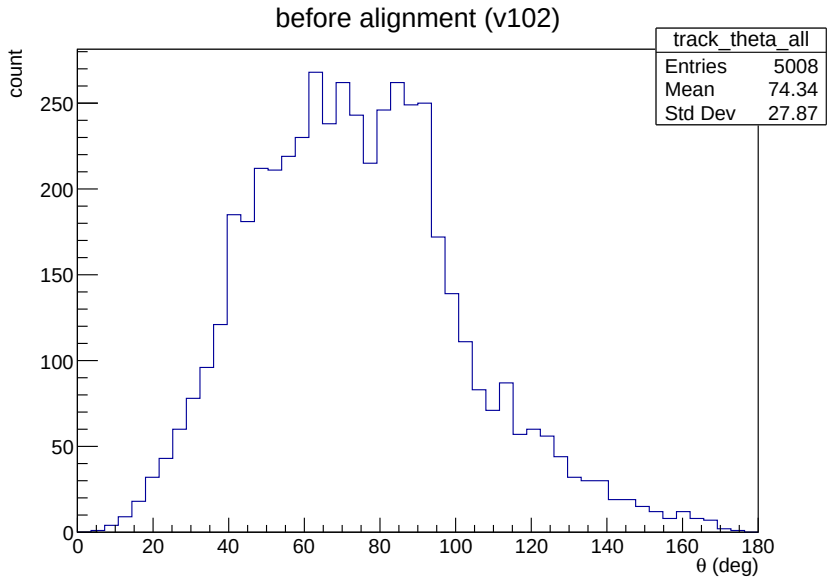
Updates - AHDC alignement

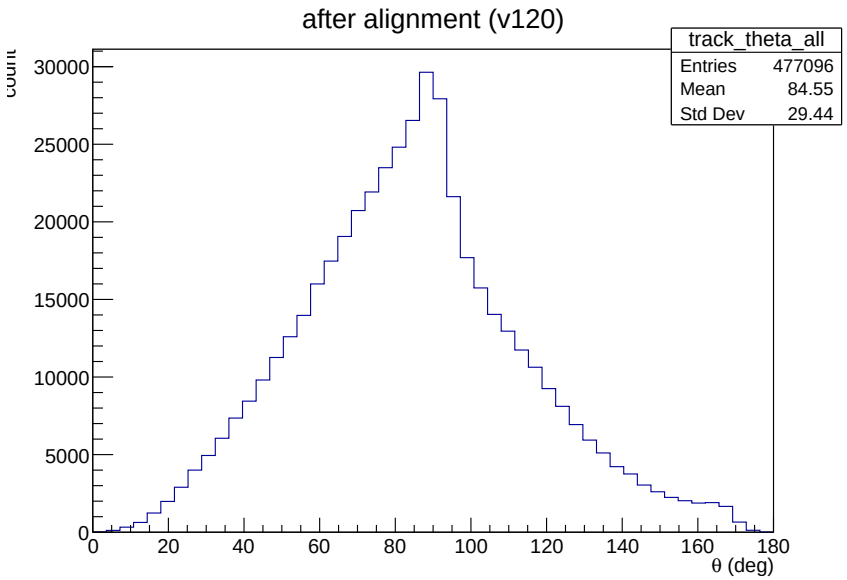
- Residual LR after **layer alignment** (first attempt, but few iterations) (no fit)
- RotationZ: 0°, +2.55°, +2.11°, 0°, -0.32°, +2.37°, +2.13°, -0.21°



Updates - AHDC alignement

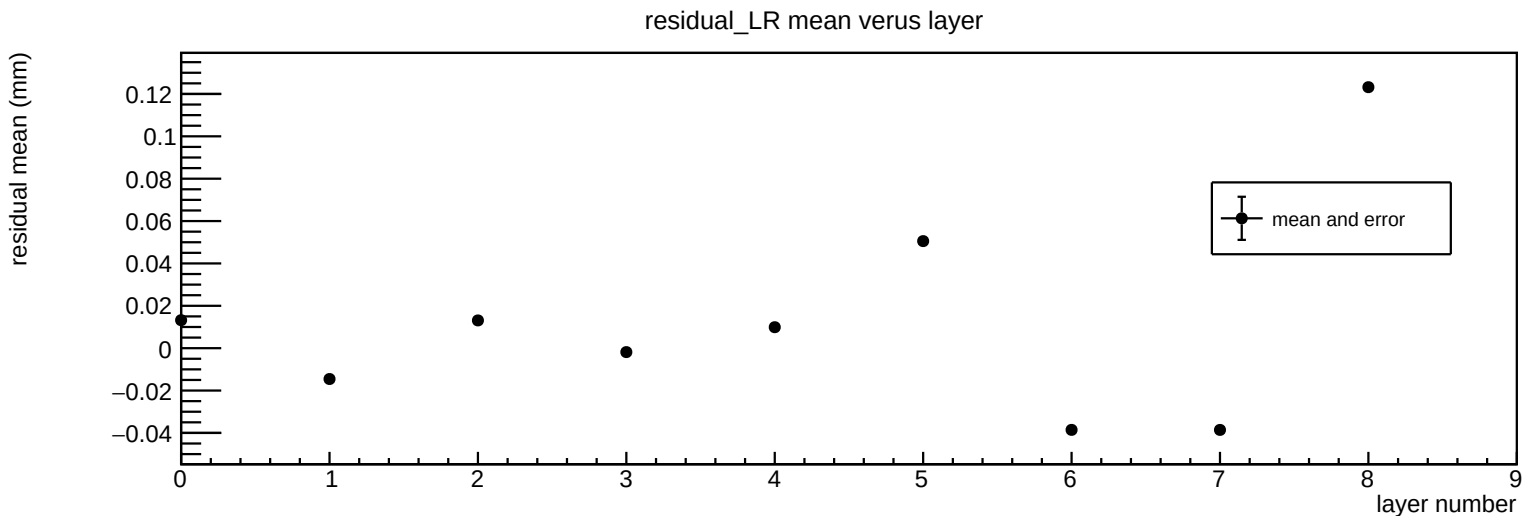
- Effect on theta (when we run the Kalman Filter fit)
- The residual_LR is a good observable for the AHDC alignment

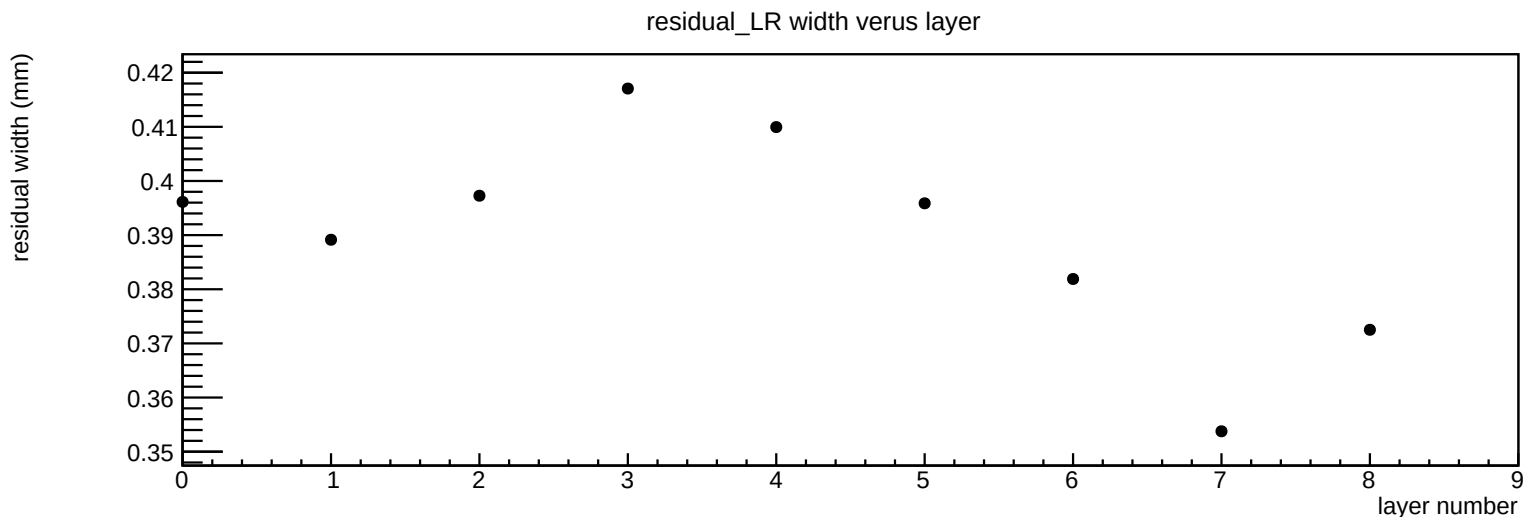




Updates - AHDC alignement

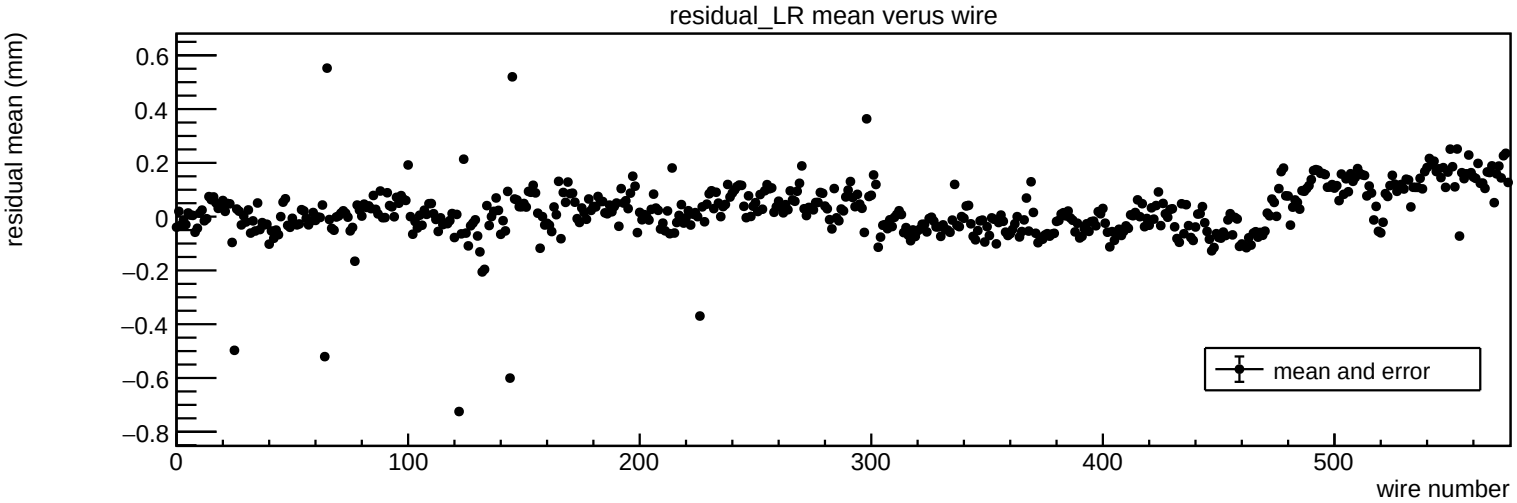
- Layer by layer residual after the fit
- After the fit, the last layer get misalign

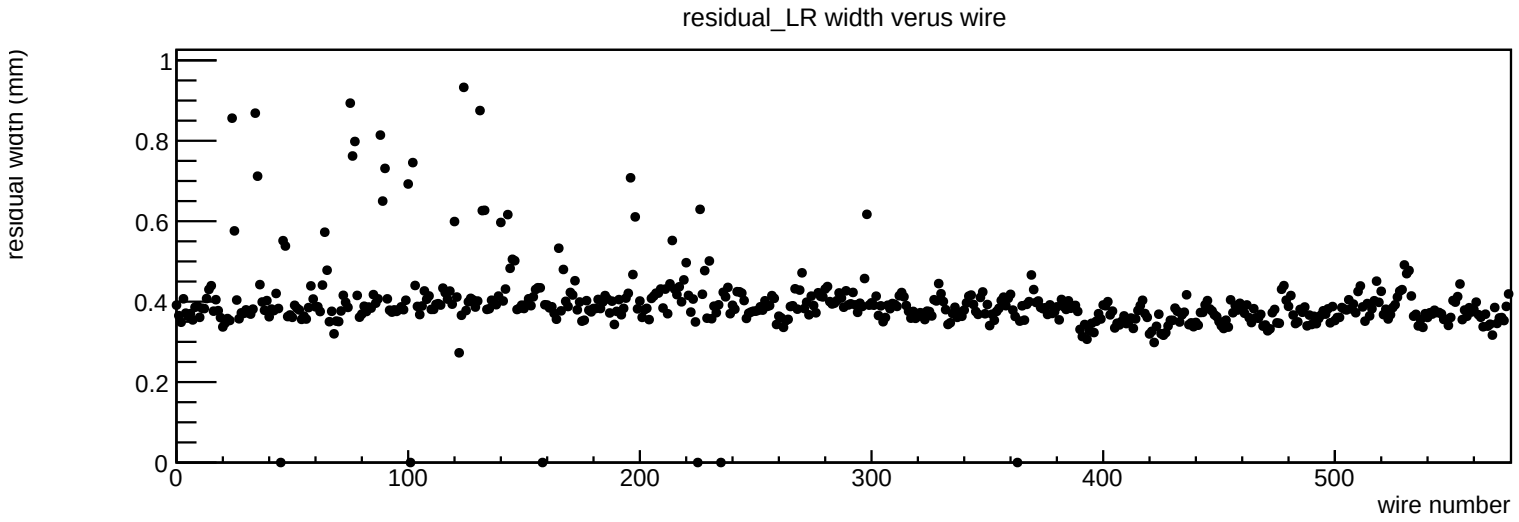


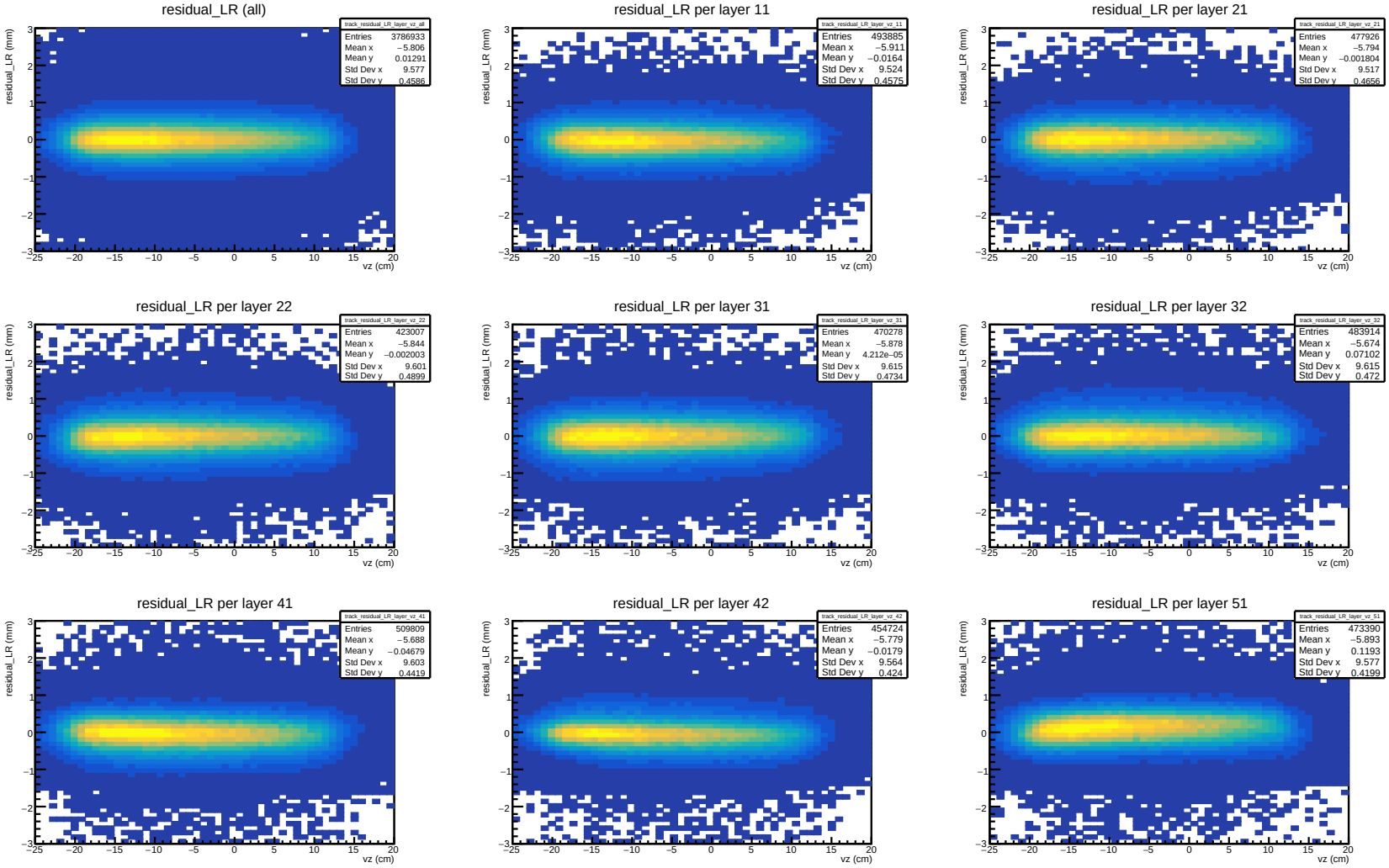


Updates - AHDC alignement

- We may need to look wire by wire (the fit is done)





Updates - AHDC alignement

- We may also have a vz dependence (the fit is done)